John Li

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Education

Northeastern University — Boston, MA
Ph.D. Computer Science — *Sep 2021 — Present*
- Advisors: Amal Ahmed and Steven Holtzen

Princeton University — Princeton, NJ
M.S.E. Computer Science — *Sep 2019 — May 2021*
- Advisor: Andrew Appel

Princeton University — Princeton, NJ
A.B. Neuroscience, High Honors — *Sep 2015 — May 2019*
- Minor in Computer Science

Work Experience

Research Intern — Jun 2023 — Present
Sandia National Laboratories — *Albuquerque, NM*
- Supervisors: Jon Aytac and Philip Johnson-Freyd

Research Intern — Jun 2021 — Aug 2021
Microsoft Research — *Redmond, WA*
- Supervisors: Tahina Ramananandro and Nikhil Swamy

Research Intern — Jun 2019 — Aug 2019
HRL Laboratories — *Malibu, CA*
- Supervisors: Aleksey Nogin and Michael Warren

Honors and Awards

ACM SIGPLAN Distinguished Paper Award — PLDI 2026

ACM SIGPLAN Distinguished Paper Award — OOPSLA 2025

Publications

1. **John M. Li**, Jack Czenszak, and Steven Holtzen. 2026. Categorical Semantics of Probabilistic Symbolic Execution. *Programming Language Design and Implementation (PLDI)*. doi:10.1145/3808343
2. Andrew Wagner, Olek Gierczak, Brianna Marshall, **John M. Li**, and Amal Ahmed. 2025. From Linearity to Borrowing. *Object-Oriented Programming Languages and Systems (OOPSLA)*. doi:10.1145/3764117
3. Sam Stites, **John M. Li**, and Steven Holtzen. 2025. Multi-Language Probabilistic Programming. *Object-Oriented Programming Languages and Systems (OOPSLA)*. doi:10.1145/3720482
4. Cameron Moy, Jack Czenszak, **John M. Li**, Brianna Marshall, and Steven Holtzen. 2025. Roulette: A Language for Expressive, Exact, and Efficient Discrete Probabilistic Programming. *Programming Language Design and Implementation (PLDI)*. doi:10.1145/3729334
5. **John M. Li**, Jon Aytac, Philip Johnson-Freyd, Amal Ahmed, and Steven Holtzen. 2024. A Nominal Approach to Probabilistic Separation Logic. *Logic in Computer Science (LICS)*. doi:10.1145/3661814.3662135
6. **John M. Li**, Amal Ahmed, and Steven Holtzen. 2023. Lilac: A Modal Separation Logic for Conditional Probability. *Programming Language Design and Implementation (PLDI)*. doi:10.1145/3591226
7. **John M. Li** and Andrew W. Appel. 2021. Deriving Efficient Program Transformations from Rewrite Rules. *International Conference on Functional Programming (ICFP)*. doi:10.1145/3473579
8. Zoe Paraskevopoulou, **John M. Li**, and Andrew W. Appel. 2021. Compositional Optimizations for CertiCoq. *International Conference on Functional Programming (ICFP)*. doi:10.1145/3473591

Teaching

Lecturer Sep 2025 — Dec 2025
Northeastern University *Boston, MA*
- Categories for PL (CS 7480)

Teaching Assistant Feb 2023 — May 2023
Northeastern University *Boston, MA*
- Logic and Computation (CS 2800), Spring 2023

Teaching Assistant Sep 2019 — Dec 2020
Princeton University *Princeton, NJ*
- Functional Programming (COS 326), Fall 2020
- Programming Languages (COS 510), Spring 2020
- Introduction to Programming Systems (COS 217), Fall 2019

Service

External Reviewer	LICS 2026
External Reviewer	CPP 2026
External Reviewer	LICS 2025
External Reviewer	POPL 2025
External Reviewer	LICS 2024

Invited Talks

Yale PL Seminar	April 2026
UIUC Logic Seminar	April 2026
Aarhus Logic Seminar	April 2025
MIT PL Review	April 2025
BU PL Seminar	July 2024
Harvard PL Seminar	October 2022